

Rajiv Gandhi University of Health Sciences, Karnataka

Fourth Semester B. Pharm Degree Examination – 18/Oct/2025

Time: Three Hours

Max. Marks: 75 Marks

Physical Pharmaceutics - II

Q.P. CODE: 5015

Your answers should be specific to the questions asked

Draw neat labeled diagrams wherever necessary

All the Questions are compulsory.

LONG ESSAYS

2 x 10 = 20 Marks

1. Discuss various properties of colloids.

OR

With relevant mathematical equation, give the construction, working and applications of cup and bob viscometer. What are its advantages?

2. What is meant by "instability" in emulsion systems? Describe various methods for the evaluation of emulsion stability.

SHORT ESSAYS

7 x 5 = 35 Marks

3. **Explain in detail, methods of purification of colloids**

OR

Describe methods to determine order of reaction.

4. What is Angle of repose? How do you determine angle of repose experimentally?

OR

Describe interfacial properties of suspended particle in a suspension.

5. Explain the plastic and pseudoplastic curves with examples. What are the reasons for such behaviour?
6. Discuss the effect of addition and removal of electrolytes in / from colloids.
7. Deduce an equation for determining the specific reaction rate constant of a first order reaction.
8. Describe air permeability method for the determination of surface area of a powder.
9. Describe different graphic presentations of size distribution data in a powder.

SHORT ANSWERS

10 x 2 = 20 Marks

10. Write two applications of micromeritics in the production of dosage forms.
11. What is Brownian Movement? Which formulations exhibit this type of movement?
12. Define 'Newtonian Flow'. Give two examples.
13. Define bulk density.
14. What is oxidation? Write any two preventive methods of oxidation.
15. Define half life of a zero order.
16. What is protective colloid?
17. What is 'Rheology'? Describe two applications of the same.
18. Explain the term 'phase inversion' with one suitable example.
19. Define suspension. Write two advantages of suspensions.
